

Experiment- 2

Metre Bridge

Aim:

To find the resistance of a given wire using a metre bridge.

Apparatus:

A metre bridge, a galvanometer, a one-way key, a resistance box, a battery, a jockey, an unknown wire resistance about one meter long, and connecting wire.

THEORY:

The unknown resistance X is given by.

$$X = \frac{(100 - l)}{l} R$$

Where R is the known resistance placed in the left gap and l (in cm) is the length of the metre bridge wire from zero and up to the balance point

PROCEDURE:

1. Arrange the apparatus as shown in the arrangement diagram
2. Connect the resistance wire whose resistance is to be determined in the right gap between C & B. Take care that no part of the wire forms a loop.
3. Connect the resistance box of the low range in the left-hand gap between A & B
4. Make all the other connections as shown in the circuit diagram.
5. Take out some resistances (say 2 ohms) from the resistance box, plug the key K
6. Touch the jockey gently first at the left end and then at the right end of the bridge wire
7. Note the deflections in the galvanometer. If the galvanometer shows deflections in opposite directions, the connections are correct. If the deflection is one side only, then there is some fault in the circuit. Rectify the fault.
8. Move (slide) the jockey gently along the wire from left to right till the galvanometer gives zero deflection. The point where the jockey is touching the wire is null point D.
9. Note length AB and call it l , BC will be $(100-l)$
10. Repeat the procedure for five different values of R . B should be such that the null point is between 40cm and 60cm of the metre bridge wire. Note the point where the galvanometer shows 0 deflection; this is called the balance point.

RESULT:

The resistance of the given wire =

Precautions

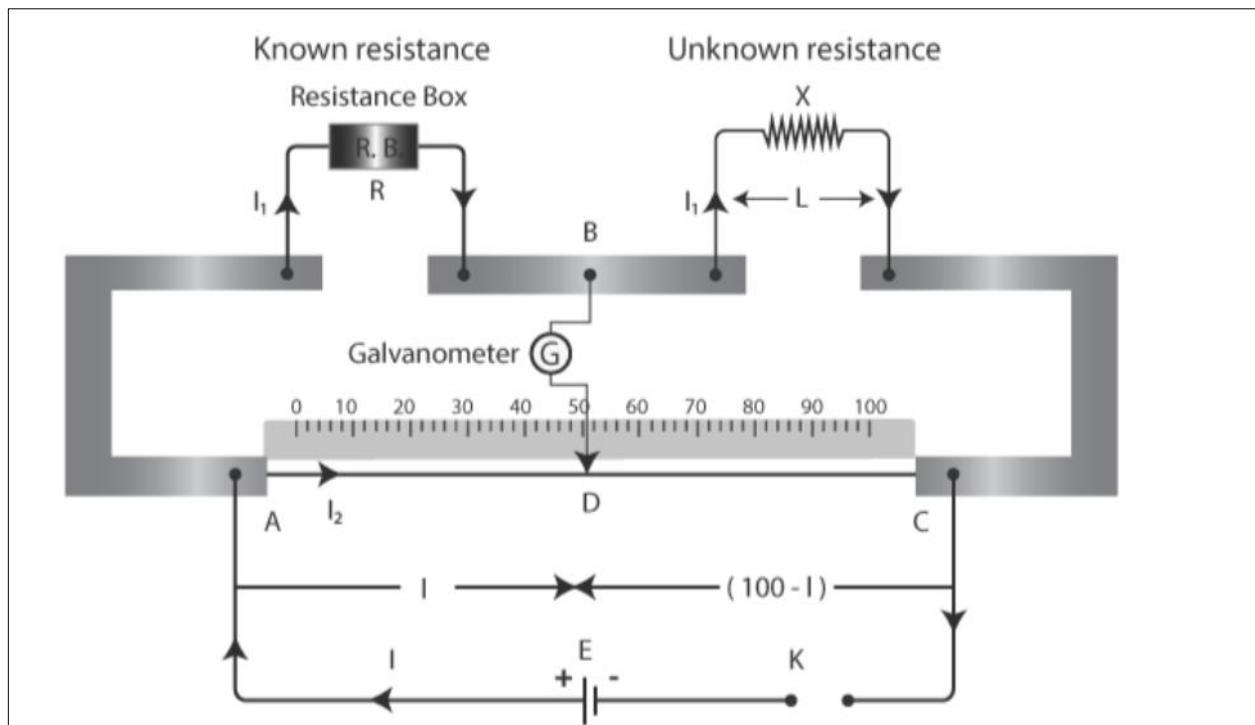
- i. All the plugs in the resistance box should be tight.
- ii. Move the jockey gently over the bridge wire and do not rub it.

Sources of Error

- i. The instrument screws may be loose.
 - ii. The wire may not have uniform thickness.
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Diagram and observation table on the left side of the page

Diagram:



Observation table

S.No.	Reading of resistance box. $R(\Omega)$	Reading of null point l (cm)	$(100-l)$ (cm)	$X = \frac{(100-l)}{l} R$ (Ω)
1				
2				
3				
4				
5				

Mean $X =$